ELE 459 Lab #4
Observer-Based Regulator for a Cart/Pendulum System

1 Introduction

The purpose of this lab is to design a digital observer-based regulator for the inverted pendulum-on-a-cart system. In addition to Matlab analysis and Simulink simulations, you will also implement the observer-based regulator to control the hardware system.

Let $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ be a continuous-time state-space model for the cart/pendulum system. Use the matrices $\mathbf{A}$ and $\mathbf{B}$ from Lab 2. The four state variables of the plant are:

$$\begin{aligned} x_1 &= \text{pendulum position (rad)} \\ x_2 &= \text{pendulum velocity (rad/sec)} \\ x_3 &= \text{motor position (rad)} \\ x_4 &= \text{motor velocity (rad/sec)} \end{aligned}$$

The two measured state variables are x_1 and x_3, so the output equation of the plant uses

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Because this system has two outputs, the $\mathbf{D}$ matrix is `D=[0;0]`. **For all of the following simulations, use the initial state vector:** `x0=[0.17;0;0;0]`.

Calculate a digital state feedback vector, $\mathbf{K}$, using the closed-loop pole locations from Lab 2 that gave the best stability margins. Record the stability margins of this full-state feedback regulator.

2 Preliminary Observer Design

An initial design of an observer is accomplished by calculating the observer gain matrix $\mathbf{L}$ using a call to `place`; namely, `L=place(phi',C',zopoles)'`. Note that this observer is a discrete-time system and so the pole locations must be inside the unit circle. **Use scaled 4th-order Bessel poles to design a 5-times faster observer for the cart/pendulum system.** Use function `dsm_regob(phi,gamma,C,K,L)` (provided on the course web site) to calculate the stability margins of this observer-based regulator. Record these margins. They are so poor that we will not try to use this regulator with the hardware, although we can see the problems caused by poor stability margins by simulating the observer-based regulator with a perturbed plant model.

The observer estimate of the plant state vector is generated by the following equation:

$$\hat{\mathbf{x}}[k+1] = (\mathbf{\Phi} - \mathbf{LC})\hat{\mathbf{x}}[k] + \mathbf{\Gamma} u[k] + \mathbf{L}\mathbf{y}[k].$$

Regulation is accomplished by feeding the estimated state vector through $-\mathbf{K}$ back to the plant input: $u[k] = -\mathbf{K}\hat{\mathbf{x}}[k]$. Substituting this $u[k]$ equation into the $\hat{\mathbf{x}}[k+1]$ equation yields a discrete-time state-update equation with input $\mathbf{y}[k]$. The output is $u[k]$ and the result is the following **observer-based regulator**:

$$\begin{aligned} \hat{\mathbf{x}}[k+1] &= (\mathbf{\Phi} - \mathbf{LC} - \mathbf{\Gamma K})\hat{\mathbf{x}}[k] + \mathbf{L}\mathbf{y}[k] \\ u[k] &= -\mathbf{K}\hat{\mathbf{x}}[k] + [0 \quad 0]\,\mathbf{y}[k]. \end{aligned}$$

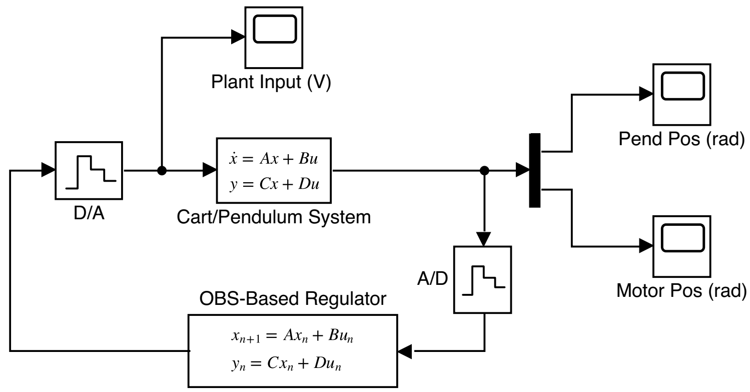

Figure 1: An observer-based regulator

Note that the `(A,B,C,D)` needed for the Simulink observer-based regulator block shown above are obtained from the equation at the bottom of page 1. The state vector of the observer should be initialized to the following value: `C\(C*x0)`. The **C** matrix used in this equation is the **plant output** matrix given at the top of page 1.

Implement an observer-based regulator as shown in Fig. 1 and simulate the response to `x0`.

The observer-based regulator should work well in simulation. However, this regulator has poor stability margins. For a given upper gain margin (UGM) in dB, the corresponding gain factor is

$$q = 10^{UGM/20}.$$

To simulate a regulator with a hardware plant that is described by a model which is different from the design model, multiply the Simulink B parameter of the Cart/Pendulum plant model by q. This multiplication should be done in the Simulink block, *not* in the Matlab m-file. Run the simulation again to see what happens. **Save this value of q to use in Part 3 of the lab. This simulation shows that an observer designed in the usual way for the cart-pendulum system results in a useless control system!**

3 Robust Observer-Based Regulator

A new method of calculating observer-gain matrices for plants having more than one measured output signal has been developed at URI [1]. The calculation of observer gains with the new m-file is accomplished as follows: `[L,delta1,delta2]=obg_reg(phi,gamma,C,K,zopoles,T)`. Notice that the function also computes the two stability robustness bounds δ_1 and δ_2. The goal is to have both of these greater than 0.5 (or at least one of them greater than 0.5).

1. Calculate `L` using `obg_reg`. Compute the stability margins using `dsm_regob` and compare them with the stability margins for full-state feedback as well as for the observer-based regulator designed in the previous section.

2. Compare the `L` matrix calculated using `place` with that calculated using `obg_reg`. Also compare `eig(phi-L*C)` for both `L` matrices.

3. Simulate the regulator with the plant input vector multiplied by the value of q obtained previously in Section 2, but now use the observer gains calculated by `obg_reg`.

References

1. R.J. Vaccaro, "An Optimization Approach to the Pole-Placement Design of Robust Linear Multivariable Control Systems," in *Proc. 2014 American Control Conference*, Portland, OR, pp. 4298-4305, June 4-6, 2014.